Supervised Learning (Part 5)

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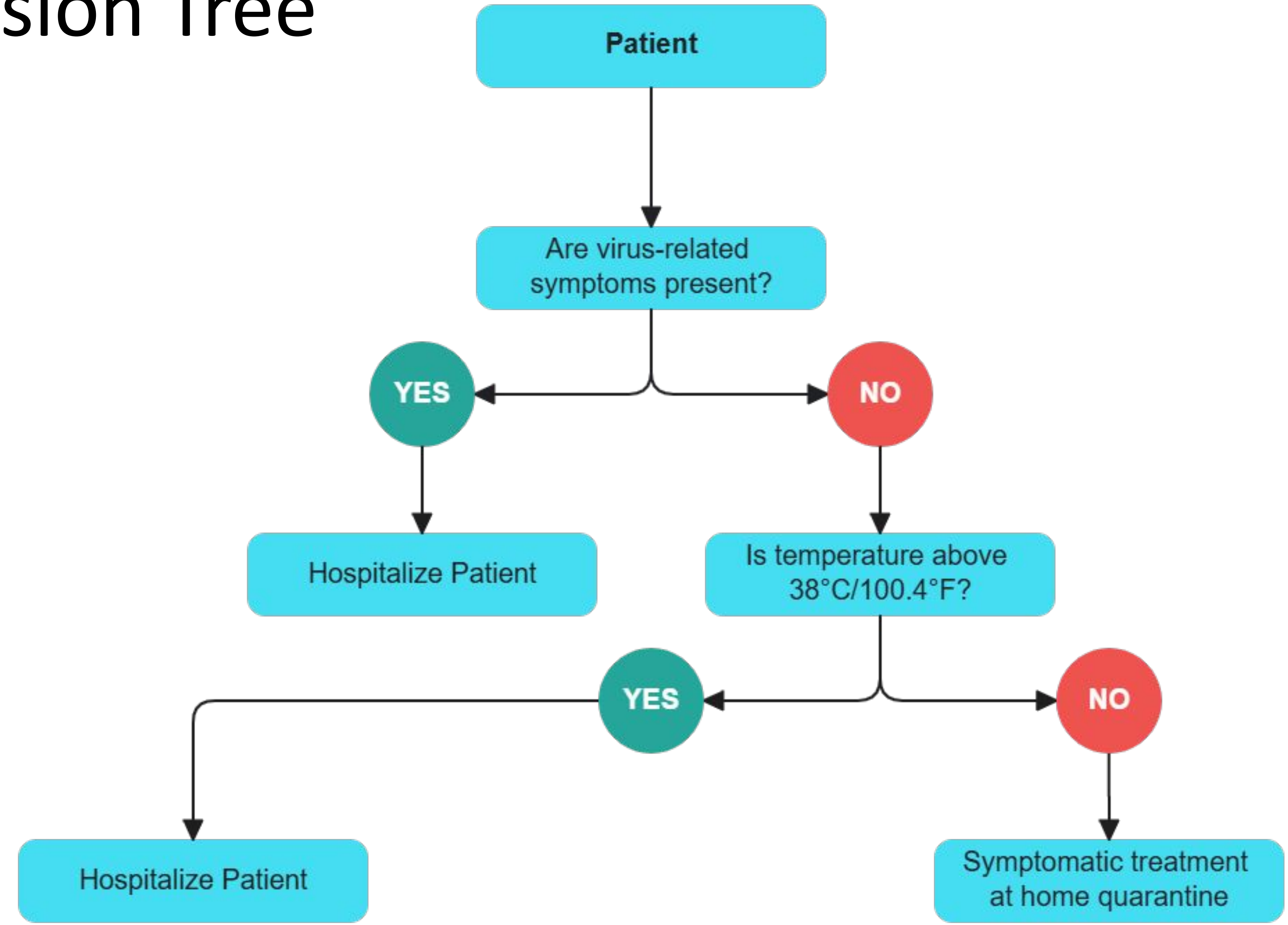
Decision Tree

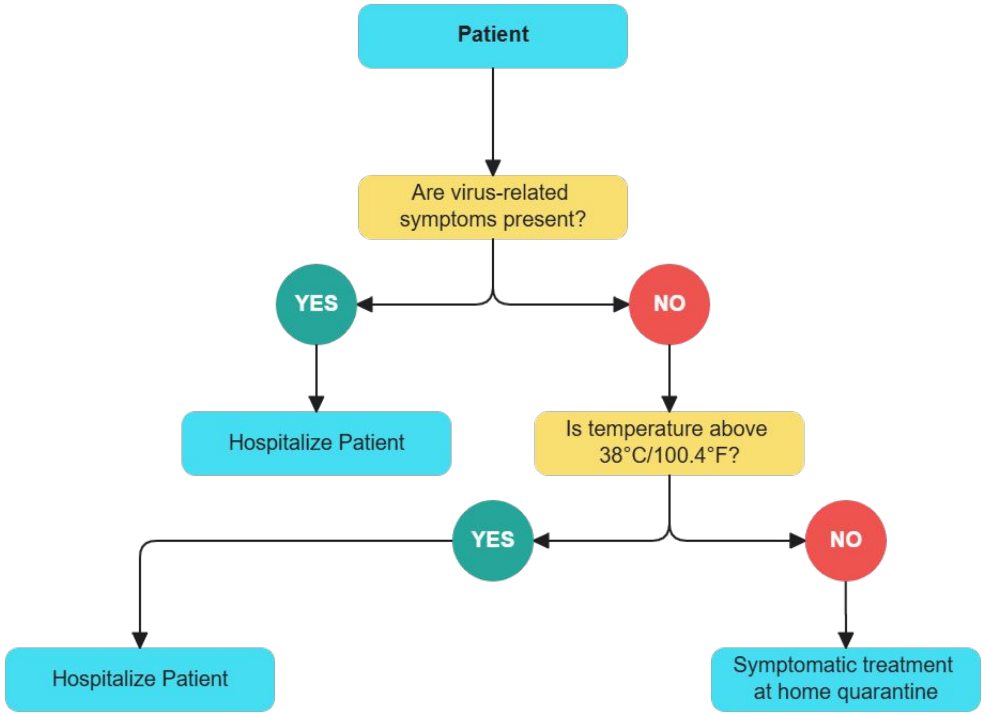
- A supervised learning algorithm
- Mimics a tree-like model of decisions and outcomes
- Used for classification and regression
- **Key Components:**
  - **Root Node**: Starting point (top of the tree)
  - **Decision Nodes**: Branches based on feature splits
  - **Leaf Nodes**: Final output (class or value)

Decision Tree



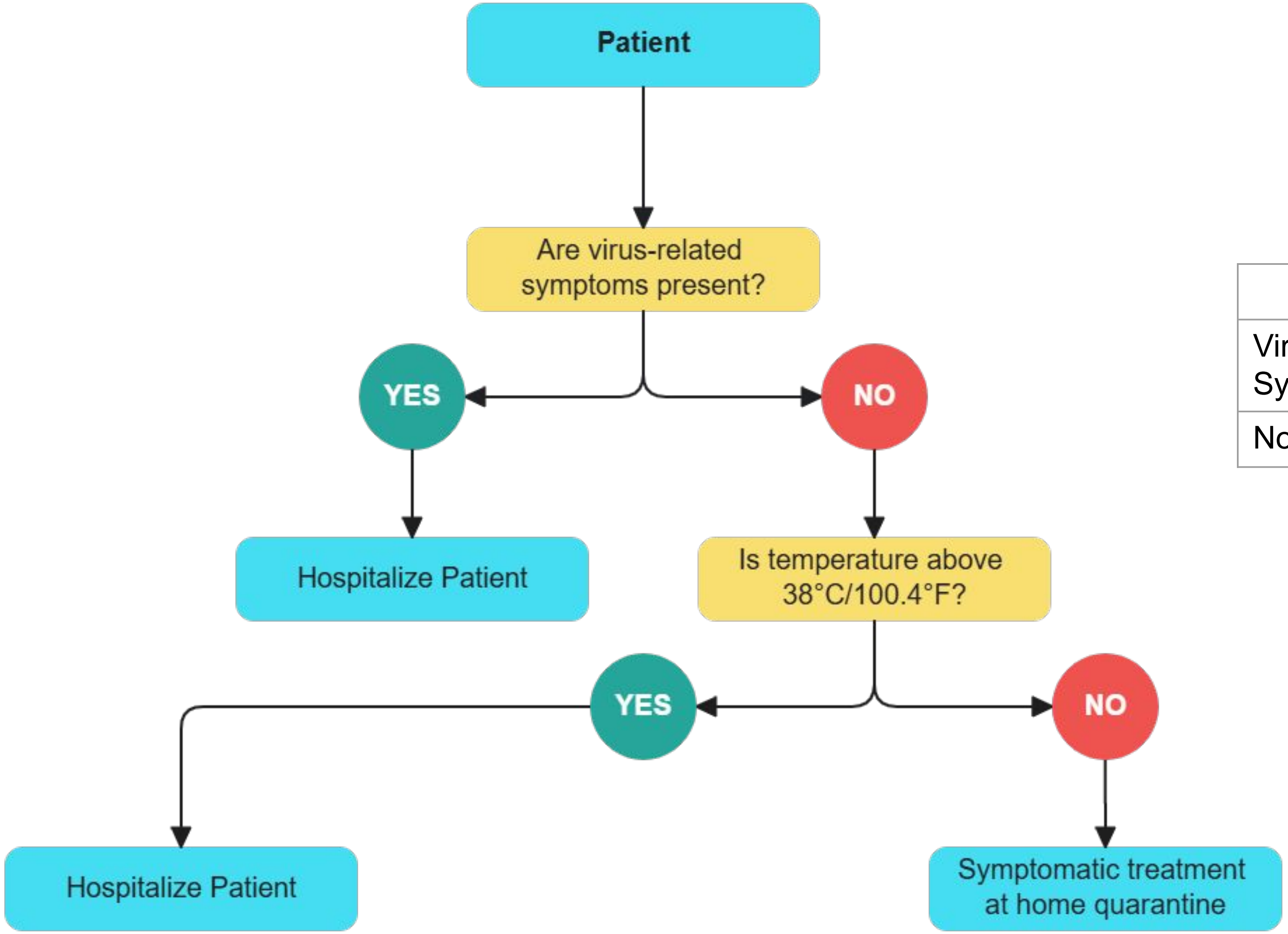
Decision Making

- Starting from the root, the attribute values are tested against the tree
- The path traced from root to leaf holds the class prediction for the tuple
- Can be found the classification rules



Viral Syndrome?	Fever?	Decision?
No	Yes	Hospitalize

Decision Making (Cont.)



Test Data		
Viral Syndrome?	Fever?	Decision?
No	No	??

Tree Construction

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1. **Split data** using features (e.g., “Temperature > 38℃?")
2. **Repeat** splitting to maximize purity (e.g., Gini impurity/Entropy for classification, MSE for regression)
3. **Stop** when a stopping criterion is met (max depth, min samples per leaf)

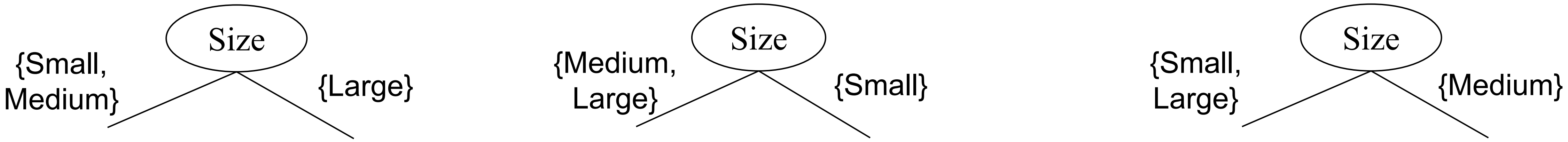
What if the Feature is not Discrete?

- Discretization to form an ordinal categorical attribute
  - Static – discretize once at the beginning
  - Dynamic – ranges can be found by equal interval for example, bucketing, equal frequency bucketing (percentiles), or clustering.

- Binary Decision: $(A < v)$ or $(A \geq v)$
  - consider all possible splits and finds the best cut
  - can be more compute intensive

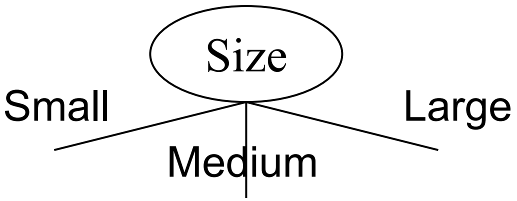
Types of Split

- **Binary split**: Divides values into two subsets



  - Need to find optimal partitioning

- **Multi-way split**: Use as many partitions as distinct values



Attribute Selection Metrics

- Information Gain
- Gain Ratio
- Gini Index

Information Gain (ID3)

- All attributes are assumed to be categorical
- Select the attribute with the highest information gain

- Entropy: A measure of uncertainty in the dataset,

  $Entropy(D) = -\sum_{i=1}^{m} p_i \log_2(p_i)$
- p_i = the number of occurrences of class *i (1…k)* divided by the total number of instances (proportion of class i)
- k = Number of class
- D = Dataset

Information Gain (ID3)

- When an attribute splits the dataset into subsets, Weighted Entropy is the average entropy of those subsets, weighted by the proportion of samples in each subset.

$$Weighted\ Entropy(A) = \sum_{j=i}^{v} \frac{|A_j|}{D} \times Entropy(A_j)$$

- Information gained by branching on attribute A

$$Gain(D, A) = Entropy(D)\ - Weighted\ Entropy(A)$$

Example Dataset

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

Example

$$Entropy(D) = E(9,5) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right) = 0.940$$

$$Weighted\ Entropy(age) = \frac{5}{14}E(2,3) + \frac{4}{14}E(4,0) + \frac{5}{14}E(3,2) = 0.694$$

$$Gain(D, age) = E(D) - WE(age) = 0.246$$

$$Gain(income) = 0.029$$

$$Gain(student) = 0.151$$

$$Gain(credit\_rating) = 0.048$$

The attribute with the maximum gain is selected as the splitting category

Example (Cont.)

Age?

<=30

Income	student	credit_rating	class
high	no	fair	no
high	no	excellent	no
medium	no	fair	no
low	yes	fair	yes
medium	yes	excellent	yes

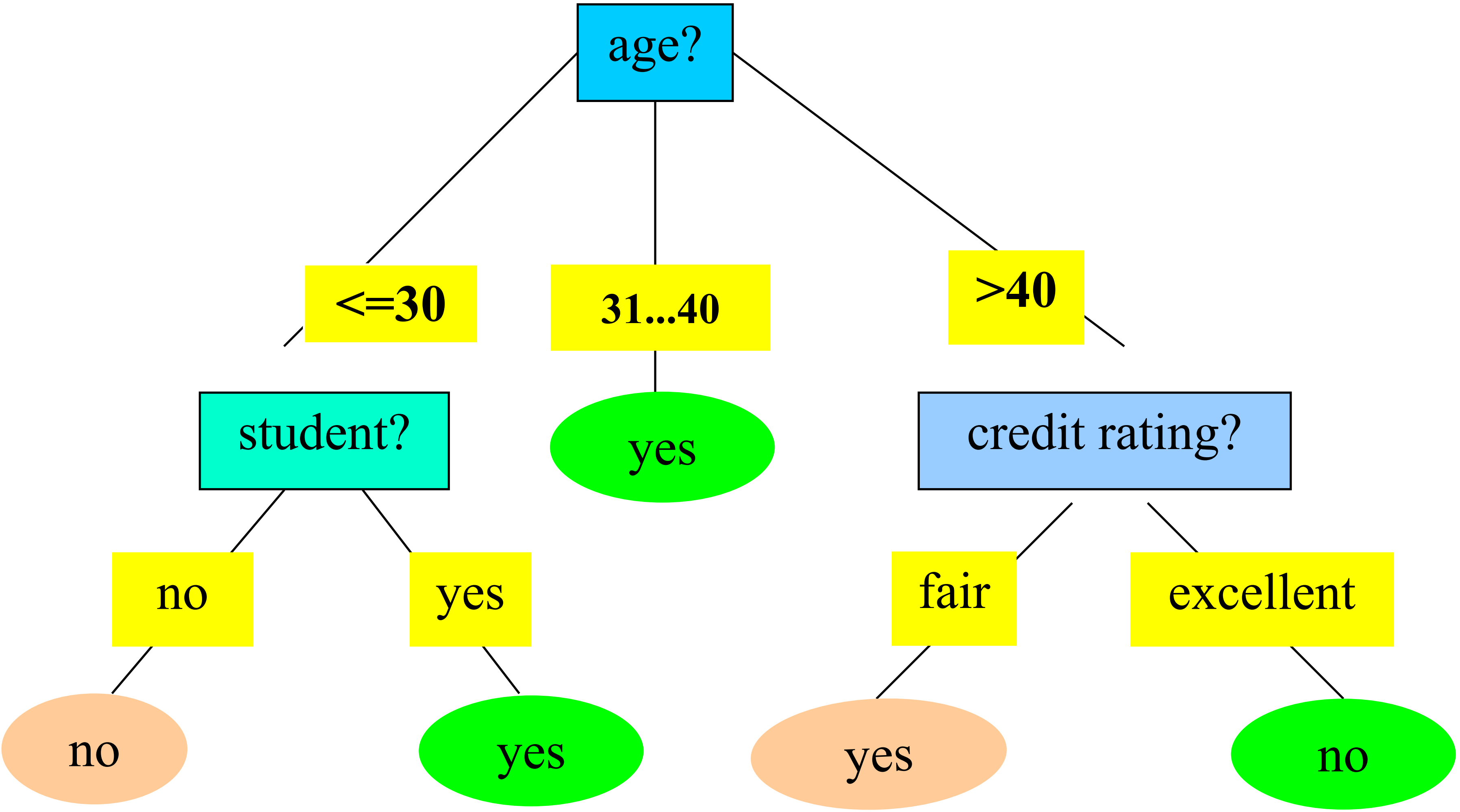
31…40

Income	student	credit_rating	class
high	no	fair	yes
low	yes	excellent	yes
medium	no	excellent	yes
high	yes	fair	yes

>40

Income	student	credit_rating	class
medium	no	fair	yes
low	yes	fair	yes
low	yes	excellent	no
medium	yes	fair	yes
medium	no	excellent	no

Example (Cont.)



Problems with Information Gain

- Information gain measure is biased towards attributes with a large number of values

- C4.5 (a successor of ID3) uses gain ratio to overcome the problem (normalization to information gain)

Gain Ratio (C4.5)

$$SplitInfo_A(D) = -\sum_{j=1}^{v} \frac{|D_j|}{|D|} \times \log_2\left(\frac{|D_j|}{|D|}\right)$$

$$GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)}$$

The attribute with the maximum gain ratio is selected as the splitting attribute

Problems with Gain Ratio

- Disadvantage of C4.5-Gain Ratio
  - Tend to prefer unbalanced split in which one partition is much smaller than the other.
- CART (Classification & Regression Tree)
  - CART creates a Binary Tree
  - Implements Gini Index as a attribute selection measure.

Gini Index

- If a data set D contains examples from n classes, gini index, gini(D) is defined as

$$gini(D) = 1 - \sum_{i=1}^{m} Pi^{2}$$

- where p_i is the relative frequency of class j in D
- If a dataset D is split on A into two subset D1 and D2, the gini index, gini(D) is defined as

$$gini_A(D) = \frac{|D_1|}{|D|} gini(D_1) + \frac{|D_2|}{|D|} gini(D_2)$$

- The split is always binary
- Reduction in impurity is defined as

$$\Delta gini(A) = gini(D) - gini_A(D)$$

Example

$$Gini\ (D) = G(9{,}5) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.4591$$

$$Gini_{Age}(D) = \frac{9}{14} Gini\left(D_{Age \le 40}\right) + \frac{5}{14} Gini\left(D_{Age > 40}\right)$$
$$= \frac{9}{14} G(6{,}3) + \frac{5}{14} G(3{,}2) = DIY$$

$$\Delta Gini(Age) = Gini(D) - Gini_{Age}(D)$$

The attribute with the Minimum Gini Index is selected as the splitting attribute

Practice Problem

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Questions?